Performance benchmark & tech stack justification for AI Image Generation project using Pollinations.ai

Executive Summary

This document presents a data-driven stack selection for building a high-performance AI image generation web app. Based on real-world benchmarks from TechEmpower Round 23 (2025), Tushar Agrawal's Go vs Python benchmark (2026), and multiple database performance studies, the recommended stack is **Go (Fiber) + React 19 + PostgreSQL** — optimised for raw throughput, low latency under concurrent load, and long-term maintainability.

102k	35k	4.5k
Go Fiber RPS	Node.js RPS	Django RPS

Benchmark: Simple JSON API, 1000 concurrent users — tusharagrawal.in (Jan 2026)

1. Backend Framework Ranking

Backend selection is the most critical decision for this project. The Pollinations.ai API can take 10–30 seconds per request (per assignment brief), so the backend must handle many concurrent long-lived connections without blocking. Go's goroutine model is purpose-built for exactly this pattern.

Rank	Framework	RPS (JSON)	Latency	Memory	Concurrency Model
#1 ✓	Go — Fiber	102,000	1.0 ms	22 MB	Goroutines (M:N)
#2	Go — Gin	95,000	1.2 ms	25 MB	Goroutines (M:N)
#3	Node.js — Express	35,000	3.2 ms	65 MB	Event loop (single-thread)
#4	Python — FastAPI	12,000	8.5 ms	85 MB	Async (asyncio)
#5	Python — Django	4,500	22 ms	120 MB	Sync / WSGI

Source: tusharagrawal.in — Python vs Go Backend Benchmark (Jan 2026), 1000 concurrent users, simple JSON API.

TechEmpower Round 23 (Feb 2025) — Fortunes test with PostgreSQL

Rank	Framework	RPS (Fortunes)	Relative to baseline
#1	ASP.NET Core (C#)	609,966	36.3x
#2 ✓	Go — Fiber	338,096	20.1x
#3	Rust — Actix	320,144	19.1x
#4	Java — Spring Boot	243,639	14.5x
#5	Node.js — Express	78,136	4.7x

Source: dev.to/tuananhpham — TechEmpower Round 23 Popular Backend Frameworks Benchmark (Feb 2025). Fortunes test = full HTML render + DB query + ORM, Postgres.

Why Go for this project

Go Fiber ranks **#2 globally** in TechEmpower Round 23 among popular frameworks, and leads all frameworks in the focused JSON API benchmark. For an AI image app where the backend must proxy slow Pollinations.ai calls while simultaneously serving gallery reads to multiple users, Go's goroutine concurrency model handles thousands of in-flight requests on minimal memory (22 MB baseline vs 65 MB for Node.js). Go also compiles to a single static binary — zero runtime dependency, trivially containerised.

2. Database Ranking

The gallery must persist across page refreshes and serve concurrent reads. Images will be stored on disk (file system / object storage), with the database holding metadata: prompt, file path, user ID, timestamp. This is a structured, relational workload — PostgreSQL is the clear fit.

Rank	Database	Read ops/s	Write ops/s	Best for
#1	Redis	100,000+	100,000+	In-memory cache / sessions
#2	MongoDB	~80,000	~80,000	Document store, flexible schema
#3 ✓	PostgreSQL	16,000	16,000	Relational, ACID, JSON support
#4	MySQL	10,000	10,000	Relational, wide hosting support
#5	SQLite	~3,000	~3,000	Embedded / single-process

Source: medium.com/@vicky542011 — Comprehensive Database Performance Comparison SQL vs NoSQL (Dec 2025), YCSB benchmark. Redis tested as in-memory store (non-persistent).

Source: dohost.us — Real-World Benchmarks: Database Performance Before and After Redis (Mar 2026).

Why PostgreSQL for this project

Redis is faster, but it's an in-memory key-value store — not suitable as the primary persistent database for gallery data. MongoDB's flexible schema has no advantage here since the data model is fixed (prompt, path, timestamp, user). **PostgreSQL at 16k ops/s is more than sufficient** for an image gallery (typical traffic is tens to hundreds of reads/sec, not thousands). PostgreSQL also offers native JSONB columns — useful for storing generation parameters and model metadata from Pollinations.ai responses — plus full ACID compliance and excellent Go driver support (pgx).

Bonus: PostgreSQL outperforms MySQL by 1.8x on write-heavy OLTP workloads (16,000 vs 10,000 ops/sec) per the same YCSB benchmark.

3. Frontend Framework Ranking

The frontend must display a gallery, show a meaningful loading state (10–30s), handle re-generation from saved prompts, and show error states. React's ecosystem dominance makes it the practical choice for this assignment.

Rank	Framework	FCP	Bundle (gz)	Developer usage (2025)
#1 perf	Svelte 5	800 ms	~15 KB	6.9%
#2 perf	Vue 4	1,000 ms	~34 KB	18.4%
#3 perf	Next.js 16	1,100 ms	~48 KB	21.5%
#4 ✓	React 19	1,200 ms	~42 KB	46.9%
#5 perf	Angular 19	1,400 ms	~130 KB	19.8%

Source: usama.codes — Svelte 5 vs React 19 vs Vue 4 (Dec 2025). FCP = First Contentful Paint, lower is better.

Source: Stack Overflow Developer Survey 2025, 49,000+ respondents — developer usage %.

Source: calmops.com — JavaScript Framework Comparison 2025 (Dec 2025) — bundle sizes gzipped.

Why React for this project

Svelte is faster (800ms FCP vs 1200ms), but React's 200ms FCP gap is imperceptible when the dominant UX wait is the 10–30s Pollinations.ai generation time. React is used by **46.9% of professional developers** (Stack Overflow 2025) — meaning easier hiring, more libraries, and more StackOverflow answers. For this assignment specifically, React's component model maps cleanly to the gallery card, loading spinner, prompt editor, and error state components described in the brief. react-query handles polling and caching the generation status elegantly.

4. Full Stack Architecture for Pollinations.ai

Request journey

1. User enters a prompt in the React frontend and hits Generate.
2. React POSTs to **Go backend** /api/generate — returns a job_id immediately (non-blocking).
3. Go spawns a goroutine that calls Pollinations.ai (https://image.pollinations.ai/prompt/{text}) — waits up to 60s.

4. On success, Go saves the image binary to /data/images/{uuid}.png on disk and inserts a row into PostgreSQL (prompt, file_path, created_at, status).
5. React polls GET /api/jobs/{job_id} every 2s until status = done or error.
6. Gallery page calls GET /api/gallery — PostgreSQL query, returns JSON list.
7. Images are served as static files from /data/images/ via Go's built-in file server.

Failure states handled

Failure	Detection	User feedback
Pollinations timeout	Go: 60s context.WithTimeout	Error card: "Generation timed out, try again"
Invalid prompt (429/4xx)	Go: HTTP status check	Error card: "Prompt rejected by API"
Broken/empty image	Go: content-length + PNG header check	Error card: "Invalid image received"

5. Sources & References

Backend benchmarks

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2. dev.to/tuananhpham — "Best popular backend frameworks by performance" — TechEmpower Round 23 (Feb 2025)
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3. jhkinfotech.com — "Best Backend Frameworks for Web Development in 2025" (Nov 2025)
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Frontend benchmarks

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URL: https://usama.codes/blog/svelte-5-vs-react-19-vs-vue-4-comparison

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